Topic: Intervals of validity

Question: Which differential equation has a solution with an interval of validity of $(-\infty, \infty)$?

Answer choices:

A $e^{-x}\dfrac{dy}{dx} - yx^3 - x^2e^{-x} + 2e^x = 0$

B $x\dfrac{dy}{dx} - (x^2 + 1)y = 3e^x$

C $y' + e^{2x}(x - 2)y - \dfrac{1}{x} = x^2$

D $x^2y' - xy\ln x = e^x$

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Solution: A

Rewrite the first equation.

$$e^{-x}\frac{dy}{dx} - yx^3 - x^2e^{-x} + 2e^x = 0$$

$$e^{-x}\frac{dy}{dx} - yx^3 = x^2e^{-x} - 2e^x$$

$$\frac{dy}{dx} - yx^3e^x = x^2 - 2e^{2x}$$

When the equation is linear, we can find the interval of validity without calculating the equation's solution, because the interval of validity is determined by $P(x)$ and $Q(x)$ from the standard form of the linear equation,

$$\frac{dy}{dx} + P(x)y = Q(x)$$

In our case,

$$P(x) = -x^3e^x$$

$$Q(x) = x^2 - 2e^{2x}$$

The functions $P(x)$ and $Q(x)$ are defined everywhere, $(-\infty, \infty)$, so the interval of validity is $(-\infty, \infty)$.

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Topic: Intervals of validity

Question: Find the interval of validity for the differential equation if $y(1) = 3$.

$$(x+1)y' = xy + x^2e^{-x^2}$$

Answer choices:

A $(-\infty, -1)$

B $(0,\infty)$

C $(-1,\infty)$

D $(-\infty,2)$

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Solution: C

Put the equation into standard form of a linear equation.

$$(x+1)y' = xy + x^2e^{-x^2}$$

$$y' = \frac{xy}{x+1} + \frac{x^2e^{-x^2}}{x+1}$$

$$y' = \frac{x}{x+1}y + \frac{x^2}{x+1}e^{-x^2}$$

$$y' - \frac{x}{x+1}y = \frac{x^2}{x+1}e^{-x^2}$$

With the equation rewritten in standard form, we can identify

$$P(x) = -\frac{x}{x+1}$$

$$Q(x) = \frac{x^2}{x+1}e^{-x^2}$$

Both functions are undefined when the denominator is 0.

$$x + 1 = 0$$

$$x = -1$$

This value separates $(-\infty, \infty)$ into two possible intervals of validity, $(-\infty, -1)$ and $(-1, \infty)$. Because we're given the initial condition $y(1) = 3$, and the value $x_0 = 1$ is contained in $(-1, \infty)$, we can say that $(-1, \infty)$ is the interval of validity.

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Topic: Intervals of validity

Question: Given the general solution to $y' = y^2 x e^x$, and the initial condition $y(0) = 1$, find the actual solution and then determine its interval of validity.

$$y = \frac{1}{e^x(1-x) + C}$$

Answer choices:

A $(1,\infty)$

B $(-\infty,1)$

C $(-\infty, \infty)$

D $(-\infty,0)$

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Solution: B

To find the actual solution, we'll substitute $x = 0$ and $y = 1$ from the initial condition,

$$1 = \frac{1}{e^0(1-0) + C}$$

$$1 = \frac{1}{1 + C}$$

$$1 + C = 1$$

$$C = 0$$

and then plug this back into the general solution.

$$y = \frac{1}{e^x(1-x)}$$

The actual solution is undefined where the denominator is 0.

$$e^x(1-x) = 0$$

Because $e^x > 0$ for every x, we can divide it out to get

$$1 - x = 0$$

$$x = 1$$

This value separates $(-\infty, \infty)$ into two possible intervals of validity, $(-\infty,1)$ and $(1,\infty)$. Because the initial condition is $y(0) = 1$, and the value $x_0 = 0$ is contained in $(-\infty,1)$, we can say that $(-\infty,1)$ is the interval of validity.

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